Heat, Air Pollution, and Test Scores: What Can We Learn Across Studies?

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Motivation

- **Environmental conditions are inputs into achievement.** Alongside families, teachers, and schools, heat and air quality can affect how much students learn and how well they perform on exams.
- **Educational losses are part of the cost of environmental exposure.** Quantifying these effects helps account for benefits of cleaner air and protection from heat that extend beyond health.
- **Individual studies provide an incomplete picture.** Estimates differ across populations, exposure windows, and research designs. A systematic review and meta-analysis can summarize their magnitude, uncertainty, and variation using a common framework.

How exposure affects achievement

Mechanisms

- **Pollution:** Illness, inflammation, and impaired cognition.
  (Horvath & Borgonovi, 2022; Aguilar-Gomez et al., 2022)
- **Heat:** Physiological strain, fatigue, and reduced attention.
  (Horvath & Borgonovi, 2022; Venegas Marin et al., 2024; Heal & Park, 2016)

Two complementary channels

- **Learning period:** Disrupts attendance and knowledge acquisition.
- **Test day:** Impairs performance given accumulated knowledge.

This study

A systematic review and meta-analysis of heat, air pollution, and test scores.

- **Average effects:** What do studies find for each hazard?
- **Timing and extremes:** How do findings differ between learning-period and test-day exposure, and between linear and extreme-exposure measures?
- **Robustness:** How sensitive are conclusions to study design, estimate selection, and standardization?
- **Across hazards:** What common findings and evidence gaps emerge, and which comparisons are meaningful?

Main findings

- **Evidence base:** Pollution has more studies. In the core sample, heat studies more often use individual fixed effects; pollution studies more often control for the other hazard.
- **Average effects:** Both hazards have negative pooled estimates, in hundredths of a test-score SD per exposure SD, with substantial variation across studies.
- **Hazard and timing:** The pooled meta-regression finds no statistically clear differences between hazards or exposure windows.
- **Extreme exposure:** Pollution evidence is sparse. Heat effects are negative and significant during learning; test-day estimates are negative but imprecise.
- **Robustness:** Negative pooled estimates persist across several checks; heat magnitudes and precision are more sensitive.
- **Gaps:** Geographic coverage is uneven; joint heat–pollution effects and the role of adaptation remain unresolved.

Contributions

- **Build on earlier syntheses:** Confirm negative average effects on test scores, with generally smaller pooled estimates under our common selection and standardization framework (Prentice et al., 2025; Villalobos & Blackman, 2025; Nguyen, 2026).
- **Comparisons across hazards and timing:** Express effects on common scales and examine differences between hazards and exposure windows.
- **Sensitivity and evidence gaps:** Assess how conclusions depend on estimate selection, study design, and score normalization; identify gaps in nonlinear evidence and coverage.
- **A reusable evidence base:** Document estimates, sources, and conversion assumptions to support replication and future updates.

Background: The linear empirical model

$$Y_i = \alpha + \beta X_j + \mathbf{C}_i'\gamma + \eta_i$$

- Y_i: academic test scores; i can denote a student or an aggregate unit such as a school or district.
- X_j: exposure at location j, linked to observation i.
- Exposure windows: around the test or during the learning period.
- $\mathbf{C}_i$: controls.

Understanding Y

- Studies typically express test scores in SD units by subtracting the reference-group mean and dividing by its SD.
- This puts the effect in context: how large is the score change relative to differences among students?
- In this study, raw-score coefficients and SEs are divided by the relevant outcome SD, as reported by the study.
- However, to compare across studies, the reference SD matters.
  - For example, on the same test, a one-point loss equals **0.10 SD** if the SD is 10, but **0.20 SD** if it is 5 (Singh, 2015; Hares, Rossiter & Evans, 2026).
- **Equal effects in SD units need not represent equal changes in learning.**

Understanding X

- **Exposure measures vary:** Heat studies use temperature measures such as daily means or maxima; pollution studies use pollutant concentrations or air-quality indices.
- **Timing matters:** Test-day exposure captures conditions during assessment; learning-period exposure captures conditions while knowledge accumulates.
- **In this study:** Linear effects are expressed per 1°C for temperature and per $1\,\mu\mathrm{g}/\mathrm{m}^3$ for $\mathrm{PM}_{2.5}$.
- **Effects per exposure SD:** Compare test-score changes associated with a one-SD increase in heat or pollution.
  - These reflect both the effect per unit and how much exposure varies in each setting.

Selection bias

The challenge: Exposure is related to family resources, school quality, and location—all of which can also affect test scores.

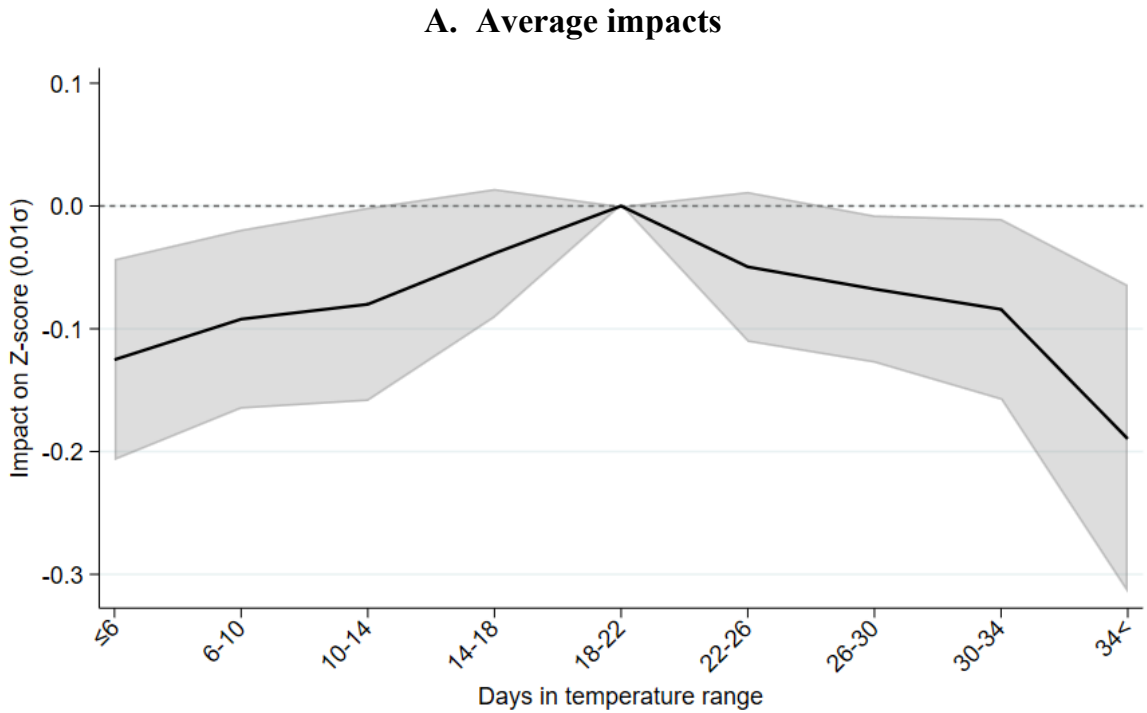
- Pollution poses additional challenges because emissions respond to economic activity; short-run temperature fluctuations are more plausibly exogenous.
- **Controls:** Adjust for observed differences in student characteristics, socioeconomic conditions, and weather.
- **Fixed effects:** Account for persistent differences across students or schools and common changes over time.
- **Instrumental variables:** Use variation that affects exposure but affects scores only through exposure.
- **Remaining concerns:** Changing student composition and test participation can still bias estimates.

Nonlinear exposure models

$$Y_{it} = \sum_{k \neq r} \kappa_k B_{jt}^k + \mathbf{C}_{it}' \boldsymbol{\psi} + \rho_i + \delta_t + u_{it}$$

- **Allow effects to vary with exposure:** Divide temperature or pollution into bins and compare each with an omitted reference bin.
- **Test-day exposure:** B_{jt}^k indicates whether exposure falls in bin k. The coefficient measures the score difference relative to the reference bin.
- **Learning-period exposure:** B_{jt}^k counts days in bin k. Holding the period length fixed, the coefficient measures the effect of replacing one reference-bin day with one day in bin k.
- **Interpretation:** These are contrasts between exposure bins, not effects per degree or per $\mu\mathrm{g}/\mathrm{m}^3$. Bin definitions and reference categories differ across studies.

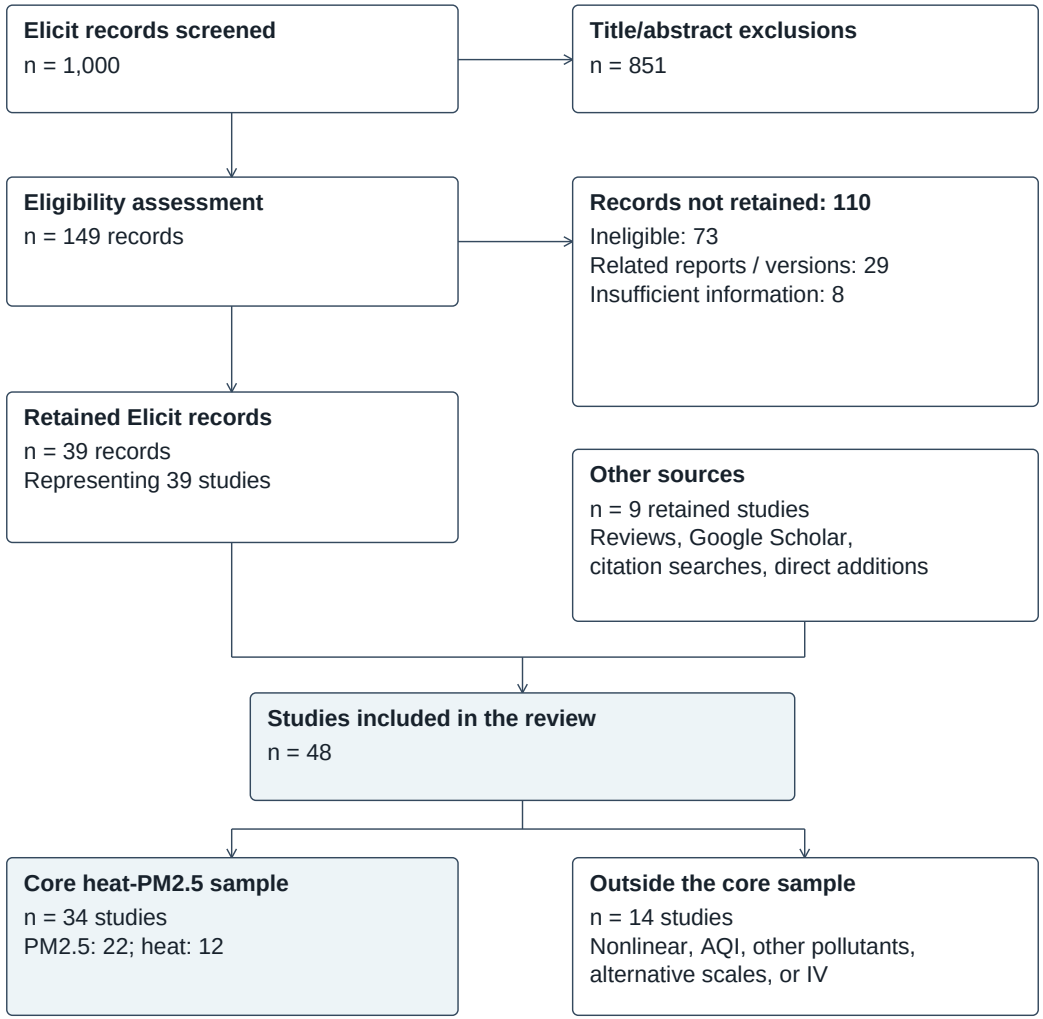
Nonlinear evidence: illustrative example



Source: Akesaka & Shigeoka (2025), Figure 3A, Japan. Daily maximum temperature on school days in the year before the test. Shaded band: 95% CI.

- **Read each bin:** One additional school day in that temperature range replaces a day at 18–22°C, the omitted reference category.
- **Read the vertical scale:** Units are 0.01 SD. An effect of −0.19 on the graph means about −0.0019 test-score SD per day.
- **Link to the synthesis:** The forest plot takes the hottest-bin contrast, here above 34°C, from each study.

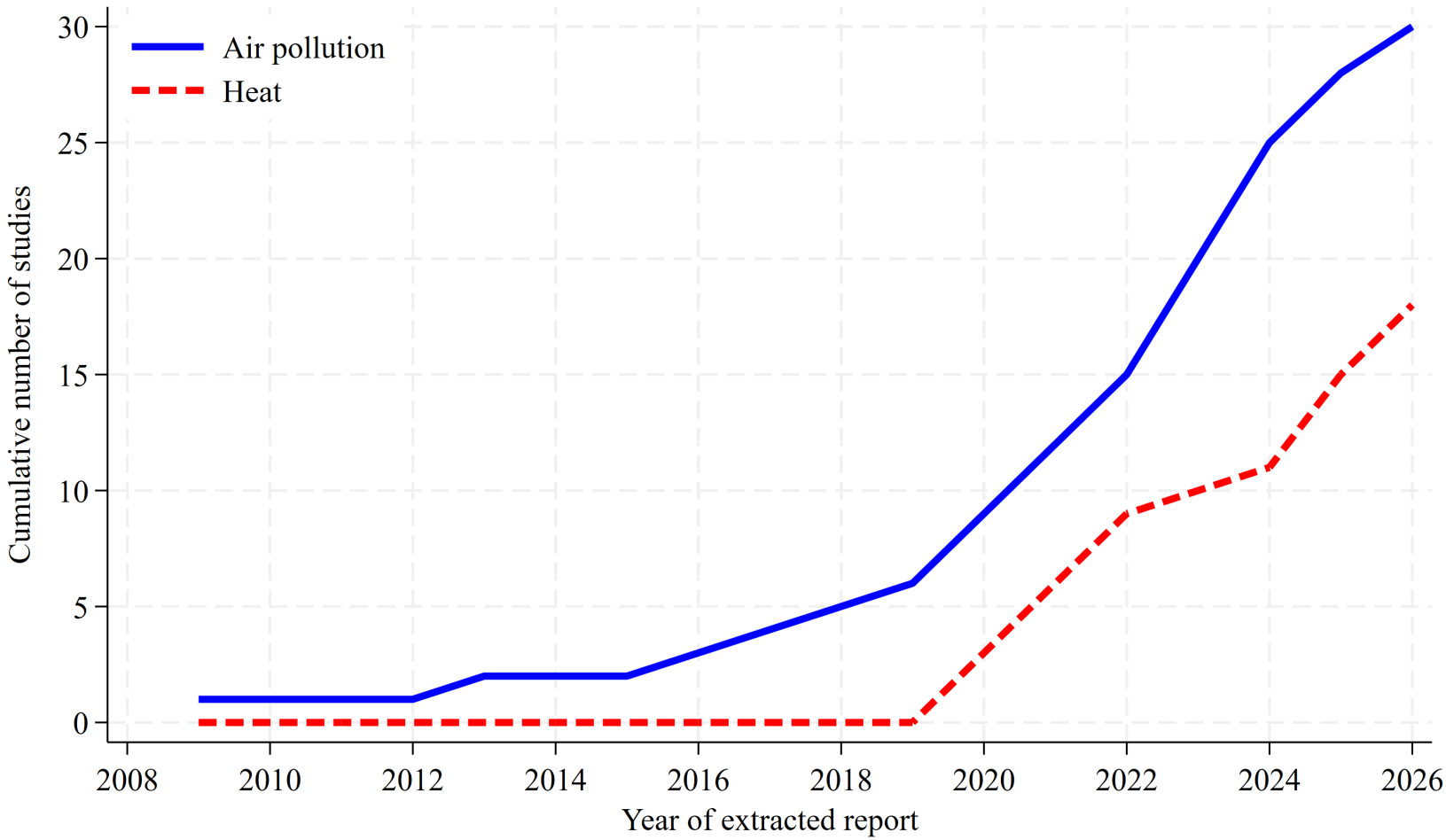
Data: Study selection



48 retained studies; 34 in the core heat–$PM_{2.5}$ sample.

Expanded standardized pool: 37 studies.

Evidence over time



Full review: 48 retained studies.

What evidence enters the main linear synthesis?

Main linear evidence	$PM_{2.5}$	Heat
Studies / estimates	22 / 89	12 / 73
Test-day (share)	0.476	0.500
Individual FE (share)	0.318	0.583
High-income country (share)	0.682	0.333
Hazard adjusted (share)	0.500	0.167

- $PM_{2.5}$ evidence is more concentrated in high-income settings.
- Controlling for the other hazard is more common in pollution studies.

Current Table 1. Shares use one selected estimate per study and nonmissing denominators. Counts include all eligible estimates. Hazard adjusted means controlling for the other hazard; it does not mean estimating an interaction.

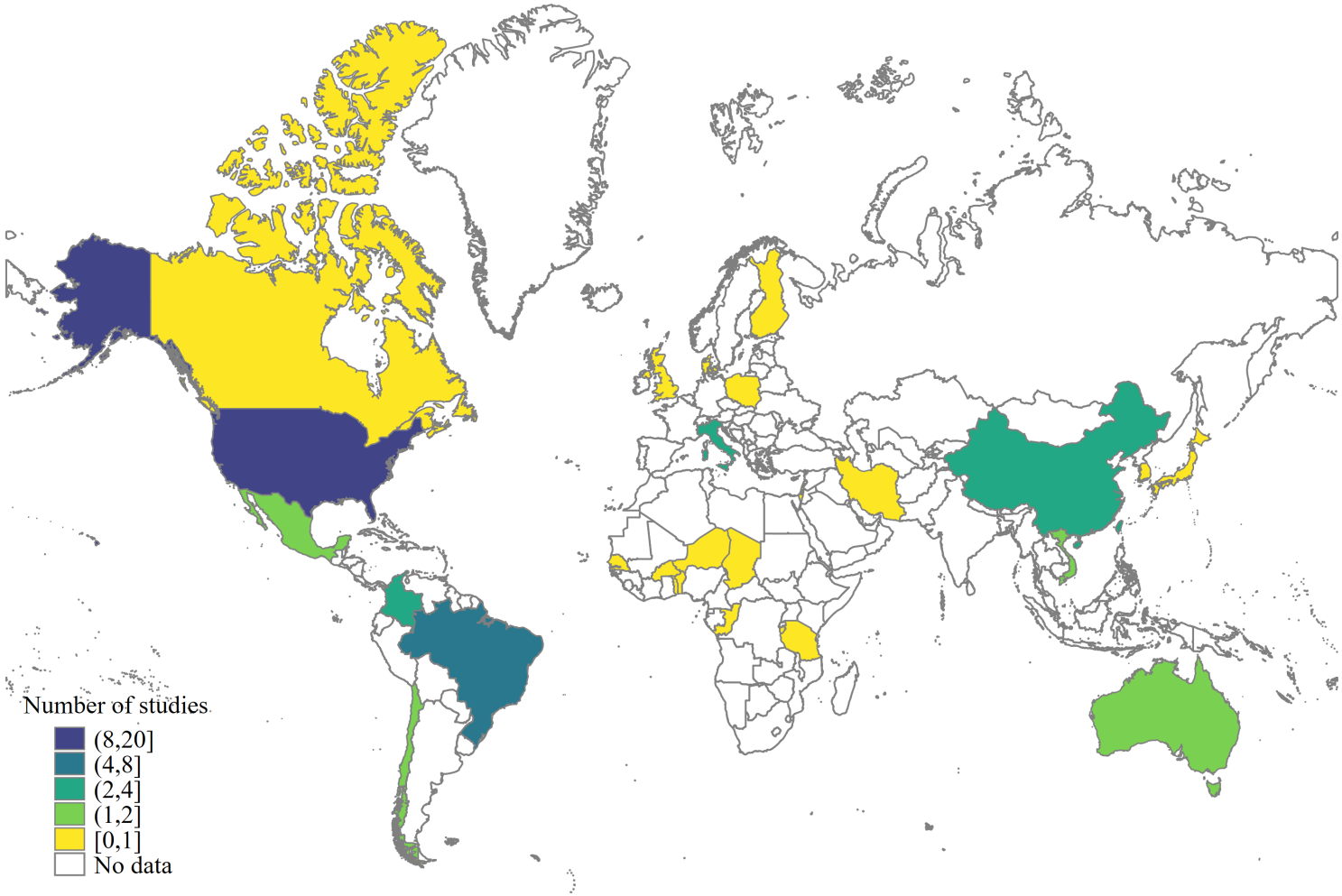


Figure 1: Studies by country

How are the results synthesized?

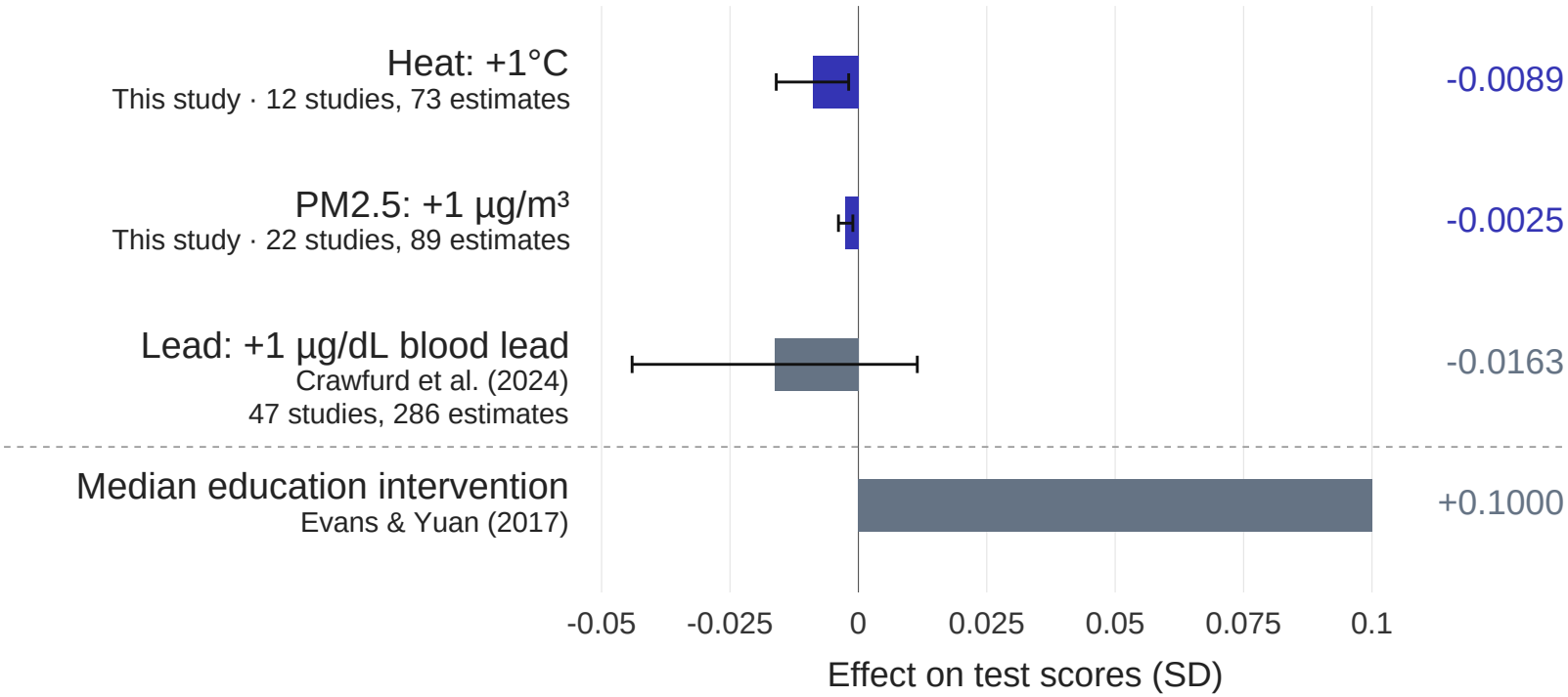
- **Main analysis:** Pool multiple estimates, accounting for dependence within studies using robust variance estimation and small-sample corrections.
- **Exposure scales:** Report heat and $PM_{2.5}$ effects per physical unit and per exposure SD. Express exposure changes in SD units to facilitate comparisons across hazards (see Hsiang, Burke & Miguel, 2013).
- **Broader coverage:** Expand the standardized pool to all eligible pollutants, including AQI, alongside heat.
- **Meta-regression:** Examine differences by hazard and exposure timing.
- **Robustness:** Compare with one preferred estimate per study and assess sensitivity to study design and score normalization.

Average effects on test scores

Hazard	Effect	95% CI	Estimates	Studies
Panel A: Per unit				
$PM_{2.5}$: 1 $\mu g/m^3$	-0.0025	[-0.0039, -0.0011]	89	22
Heat: 1°C	-0.0089	[-0.0160, -0.0019]	73	12
Panel B: Heat and $PM_{2.5}$, per exposure SD				
$PM_{2.5}$	-0.0122	[-0.0195, -0.0049]	83	20
Heat	-0.0238	[-0.0481, 0.0006]	71	11
Heat + $PM_{2.5}$	-0.0146	[-0.0215, -0.0076]	154	31
Panel C: Expanded all-hazards sample, per exposure SD				
All hazards	-0.0185	[-0.0256, -0.0115]	433	37

Notes: Effects in test-score SD. Panels A–B: multiple sample. Panel C: all eligible preferred and robustness estimates across hazards with available per-SD conversions. Random-effects robust variance estimation, clustered by study; $\rho = 0.8$, small-sample corrections; 95% CIs. Panels B–C use study-specific exposure SDs.

Illustrative effect-size benchmarks



Notes: Heat and $PM_{2.5}$: multiple-estimate effects and 95% CIs from Panel A. Lead: 95% CI calculated from the reported SE. Lead effect converted to a per-unit slope at the sample's average blood-lead level (7.06 μg/dL). **Bar lengths depend on the exposure increment chosen; they do not rank hazards or policy benefits.**

Pooled meta-regression: hazard and timing

	Coefficient	95% CI	*p*-value
Constant: pollution, test day	-0.0182	[-0.0290, -0.0075]	0.003
Heat relative to pollution	-0.0044	[-0.0276, 0.0188]	0.688
Learning period relative to test day	-0.0025	[-0.0205, 0.0155]	0.774

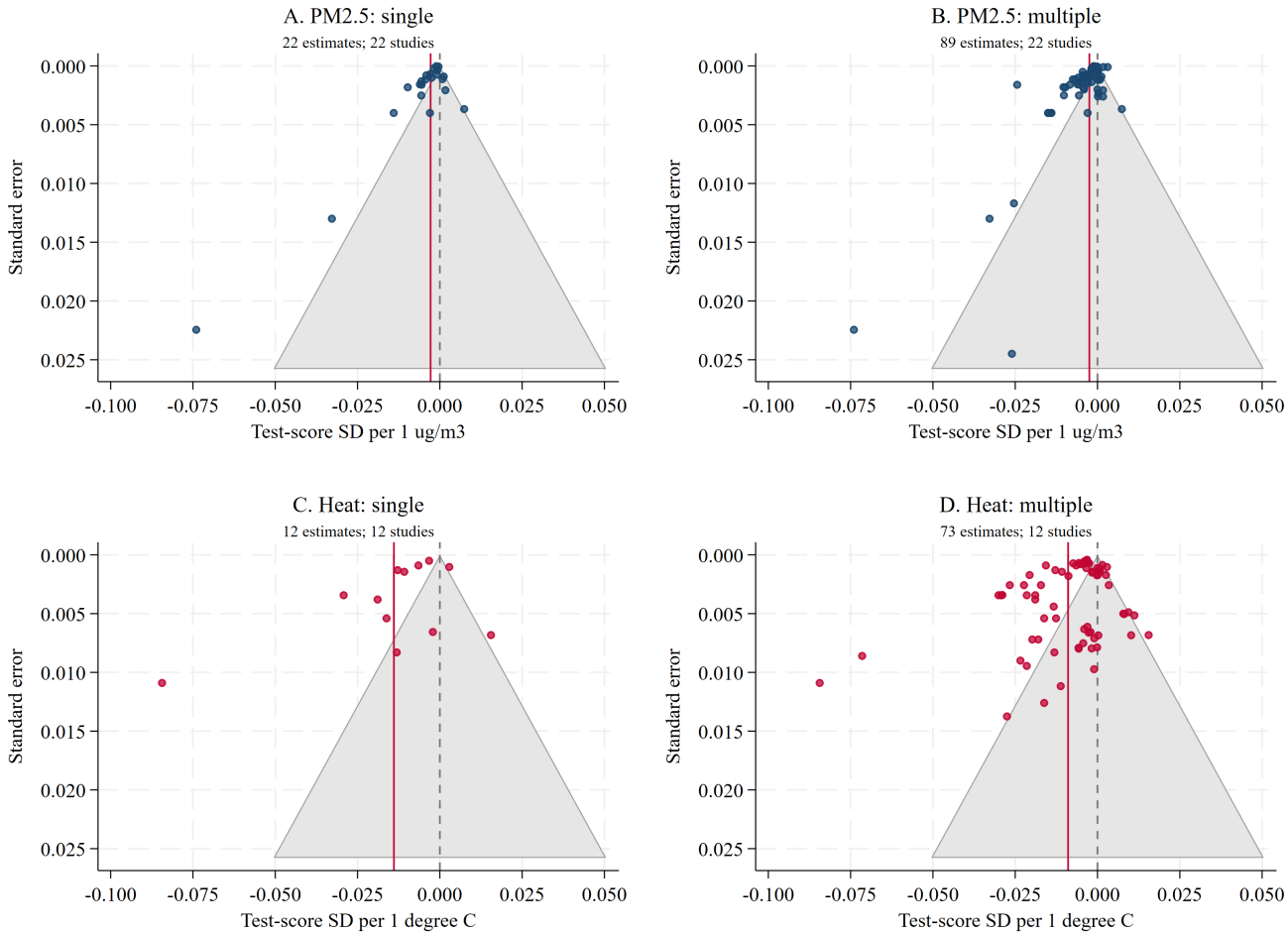
Notes: Test-score SD effect per exposure SD. Panel C sample with classified timing: 430 estimates from 36 studies. Additive model; random-effects robust variance estimation clustered by study, $\rho = 0.8$, small-sample corrections. Three estimates with unresolved timing excluded.

Neither the hazard nor the timing contrast is statistically distinguishable from zero.

Robustness

- **Estimate selection:** Pooled effects remain negative with preferred or multiple estimates; magnitudes depend on weighting.
- **Broader coverage:** Adding other pollutants preserves the negative pooled effect.
- **Study design (Table A6):** $PM_{2.5}$ findings persist under stricter fixed-effects restrictions. Heat estimates remain negative but become less precise with individual fixed effects.
- **Score normalization (Table A7):** Excluding estimates standardized by the dispersion of group-average scores preserves the $PM_{2.5}$ findings; the heat sample is unchanged.

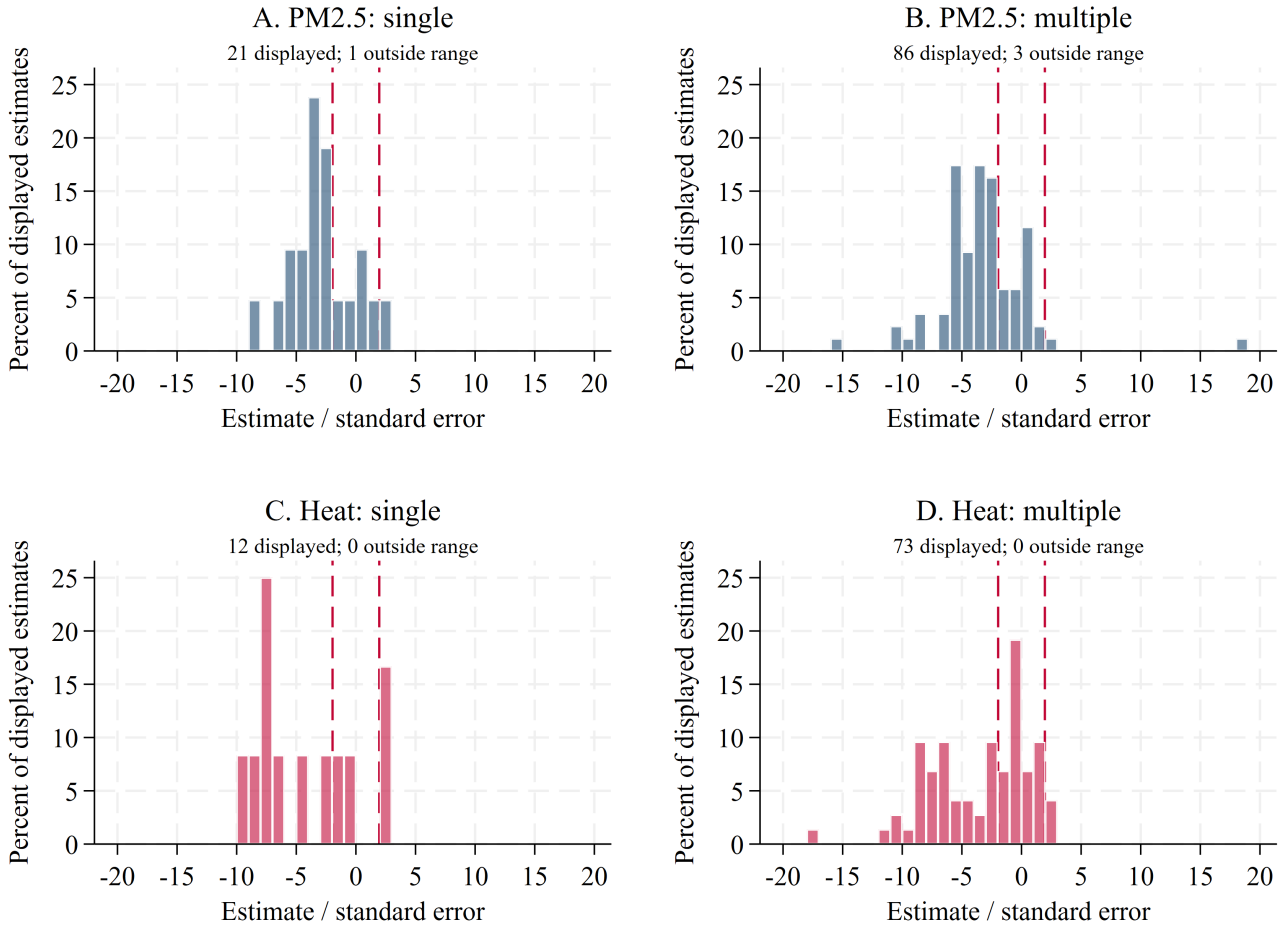
Reporting bias: what do the plots tell us?



Notes: Precision increases upward. Left: one estimate per study; right: multiple dependent estimates.

- Less precise estimates are often more negative, particularly for $PM_{2.5}$: possible overstatement of harm.
- Asymmetry may reflect study differences; it does not establish reporting bias.

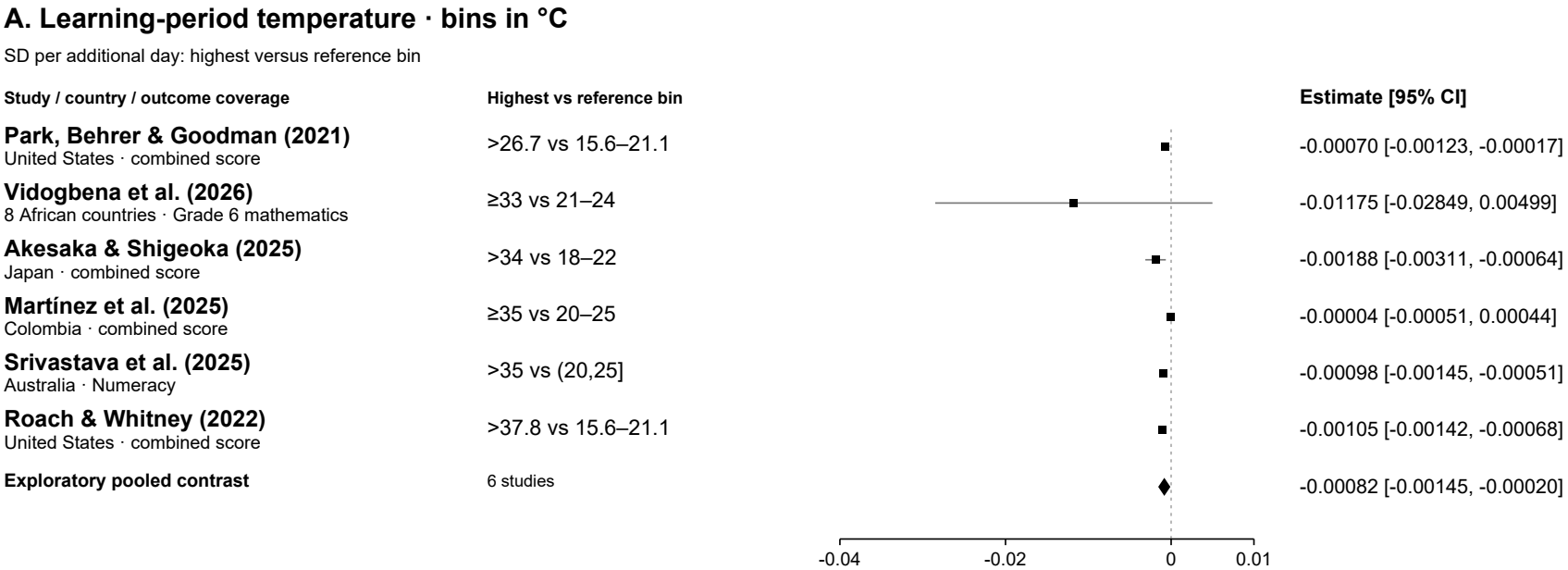
Reporting bias: significance thresholds



Notes: Left: one preferred estimate per study; right: multiple dependent estimates. Dashed lines: ± 1.96 (normal-reference 5% thresholds). Only $|\text{estimate}/\text{SE}| < 20$ shown; panel counts identify omitted values.

Some clustering among negative, significant pollution estimates; this does not establish selective reporting.

Nonlinear effects: heat during learning



Notes: Effects in test-score SD per additional day; 95% CIs. Excluding Vidogbena leaves the pooled effect essentially unchanged.

Highest versus reference temperature bin; thresholds differ across studies.

Nonlinear evidence: what do we learn?

- **Evidence gap:** Most nonlinear evidence concerns heat. Pollution evidence is sparse, especially during the learning period.
- **Heat during learning:** Ten additional days in the hottest temperature bin, replacing reference-bin days, correspond to about **0.008 SD lower scores** (95% CI: -0.015 to -0.002), assuming effects accumulate linearly.
- **Test-day heat:** Studies generally compare temperatures above 30°C (86°F) with moderate conditions, often 20–25°C (68–77°F). Four of five estimates are negative, but the pooled effect remains imprecise.
- **Across hazards:** Different thresholds and reference categories prevent ranking hazards or establishing disproportionate harm at extreme levels.

Where to go now?

- **Fill evidence gaps:** Study underrepresented settings and exposure ranges; assess where existing findings generalize.
- **Improve comparability:** Clarify dose–response relationships and which magnitude comparisons are meaningful; report exposure distributions, reference categories, and outcome SDs.
- **Broaden the comparison:** Extend the synthesis to noise, extreme rainfall, and limited access to green space, using comparable outcomes and exposure windows.
- **Explain differences:** Examine whether adaptation, school conditions, and exposure timing account for variation across studies.
- **Connect effects to policy:** Evaluate feasible exposure reductions and their costs, alongside changes in test scores.

Conclusions

- **Average effects:** Heat and air pollution are associated with lower test scores on average, with substantial variation across studies.
- **Robustness:** $PM_{2.5}$ findings persist across several checks; heat magnitudes and precision are more sensitive to selection and weighting.
- **Timing and extremes:** Timing differences remain imprecise. Extreme-exposure evidence is clearest for heat during learning.
- **Policy relevance:** Reducing harmful exposure may improve learning, but these estimates alone cannot rank hazards or interventions.

Limitations

- **Comparability:** Standardized scores and exposures do not ensure equivalent learning changes or exposure contrasts; some SD conversions rely on assumptions.
- **Generalizability:** Evidence is uneven across locations, hazards, and exposure windows.
- **Identification and reporting:** Study designs differ, and selective reporting may influence the evidence. Diagnostics do not establish its extent.
- **Unresolved mechanisms:** The synthesis cannot establish how adaptation, school conditions, or combined heat and pollution shape effects.

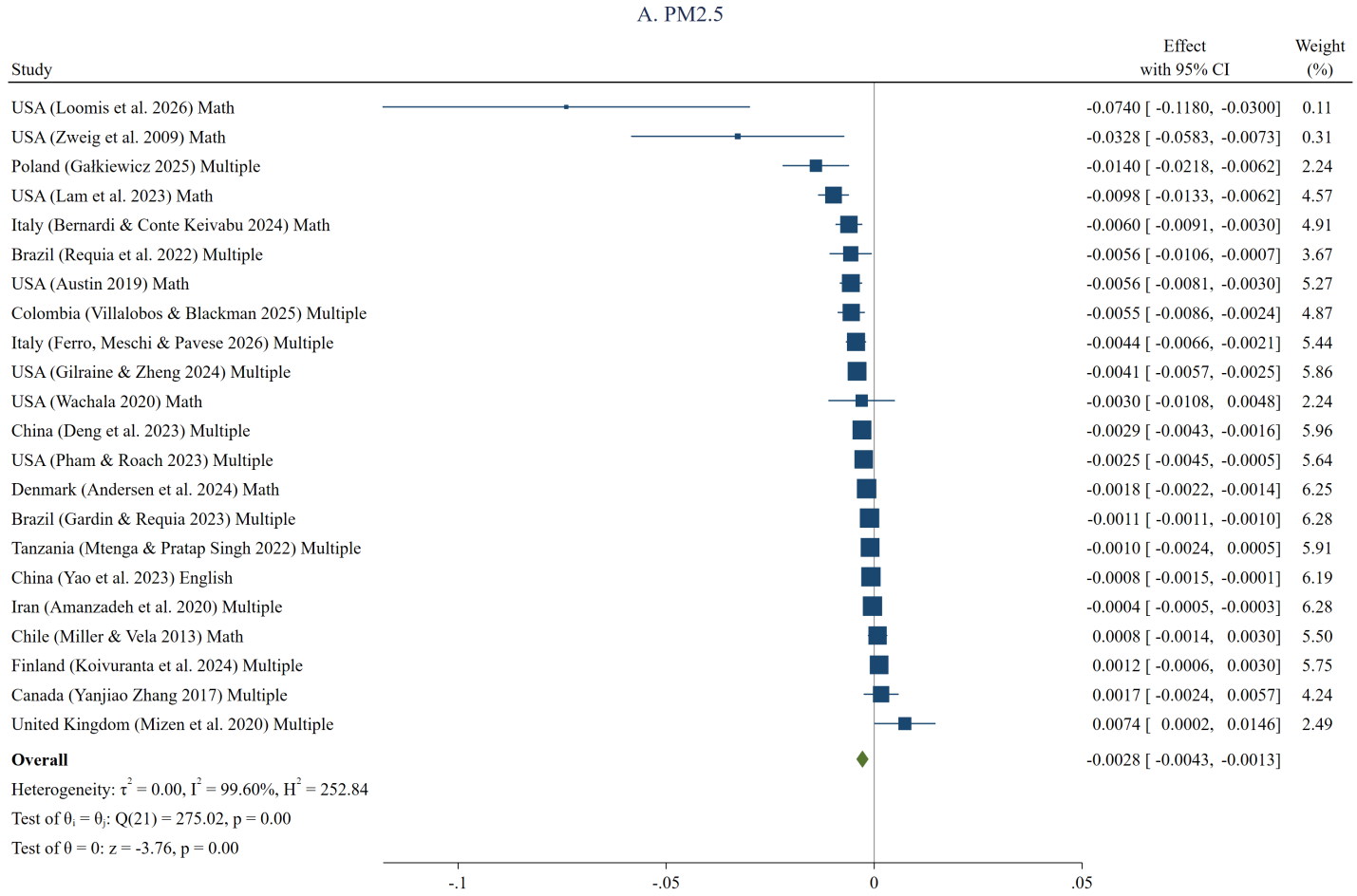
Thank you!

Questions?

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Study estimates: $PM_{2.5}$

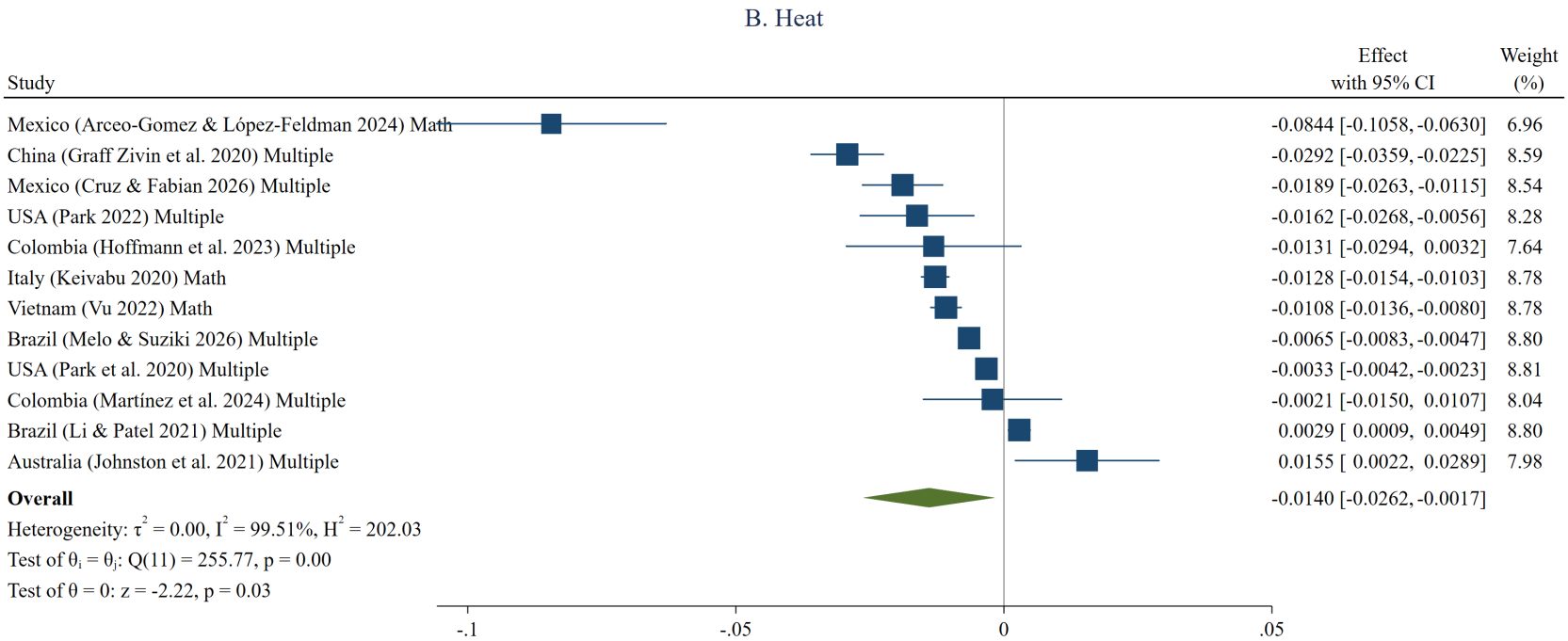
One preferred estimate per study



Random-effects REML; 95% CIs. Effects in test-score SD per 1 $\mu g/m^3$ of $PM_{2.5}$.

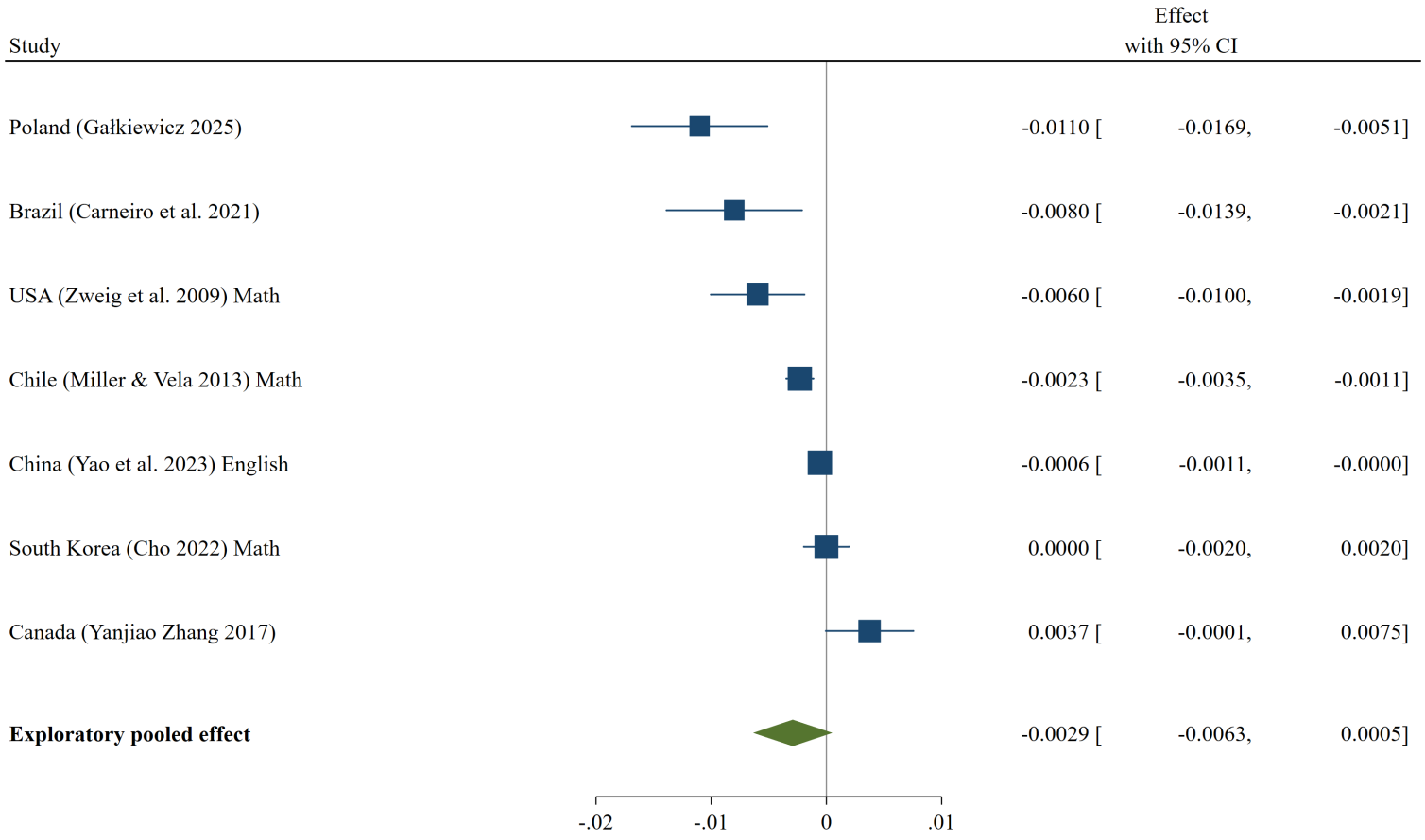
Study estimates: heat

One preferred estimate per study



Random-effects REML; 95% CIs. Effects in test-score SD per 1°C.

PM_{10}: a smaller, less precise evidence base



Seven studies: exploratory pooled effect -0.0029 score SD per $\mu g/m^3$ (95% CI: -0.0063 to 0.0005).

One selected estimate per study; exposure windows vary. Zhang uses an assumed score SD of 7.74; Cho's reported coefficient rounds to zero.

Other pollutants: evidence is limited and mixed

- **NO_2 and SO_2:** estimates are mostly negative, but several confidence intervals include zero.
- **Ozone and CO:** signs and precision vary across studies; there is no consistent pattern across all reported specifications.
- **NO_x and NO:** only a few selected estimates, with mixed statistical precision.
- **Air-quality indices:** several negative estimates, including IV results, but pollutant-specific and composite indices measure different exposures.
- **Interpretation:** retain separate pollutants and units ($\mu g/m^3$, ppb, ppm, index points). Sparse evidence and scaling qualifications limit pooling and comparisons.

Exposure-SD scaling changes the comparison

Hazard / sample	Effect	95% CI	Studies / estimates
$PM_{2.5}$: single	-0.0159	[-0.0245, -0.0073]	20 / 20
$PM_{2.5}$: multiple	-0.0122	[-0.0195, -0.0049]	20 / 83
Heat: single	-0.0474	[-0.0931, -0.0017]	11 / 11
Heat: multiple	-0.0238	[-0.0481, 0.0006]	11 / 71

Notes: Effects in test-score SD per exposure SD; 95% CIs. Single: one preferred estimate per study, random-effects REML with normal-reference intervals. Multiple: random-effects robust variance estimation, clustered by study ($\rho = 0.8$), with small-sample t corrections.

- **Heat uncertainty:** The multiple-estimate interval includes zero. For single estimates, modified Hartung–Knapp inference leaves the effect unchanged but widens the interval to $[-0.1019, 0.0071]$, also including zero (Table A4).
- **Comparability:** Exposure SDs reflect local variation, not a common physical dose; missing SDs also change the sample.

Does exposure timing explain the differences?

	$PM_{2.5}$	Heat
Learning period	-0.0041	-0.0116
95% CI	[-0.0067, -0.0016]	[-0.0297, 0.0066]
Test day	-0.0017	-0.0067
95% CI	[-0.0032, -0.0001]	[-0.0136, 0.0002]
Timing contrast	0.0020	0.0028
95% CI	[-0.0010, 0.0049]	[-0.0143, 0.0199]
Contrast p-value	0.174	0.720

Differences between windows are not statistically significant for either hazard.

Table A5, multiple estimates; per-unit effects. Contrast: test day minus learning period, estimated by meta-regression (not subtraction of the separate pools). These are associations across specifications, not causal timing effects.

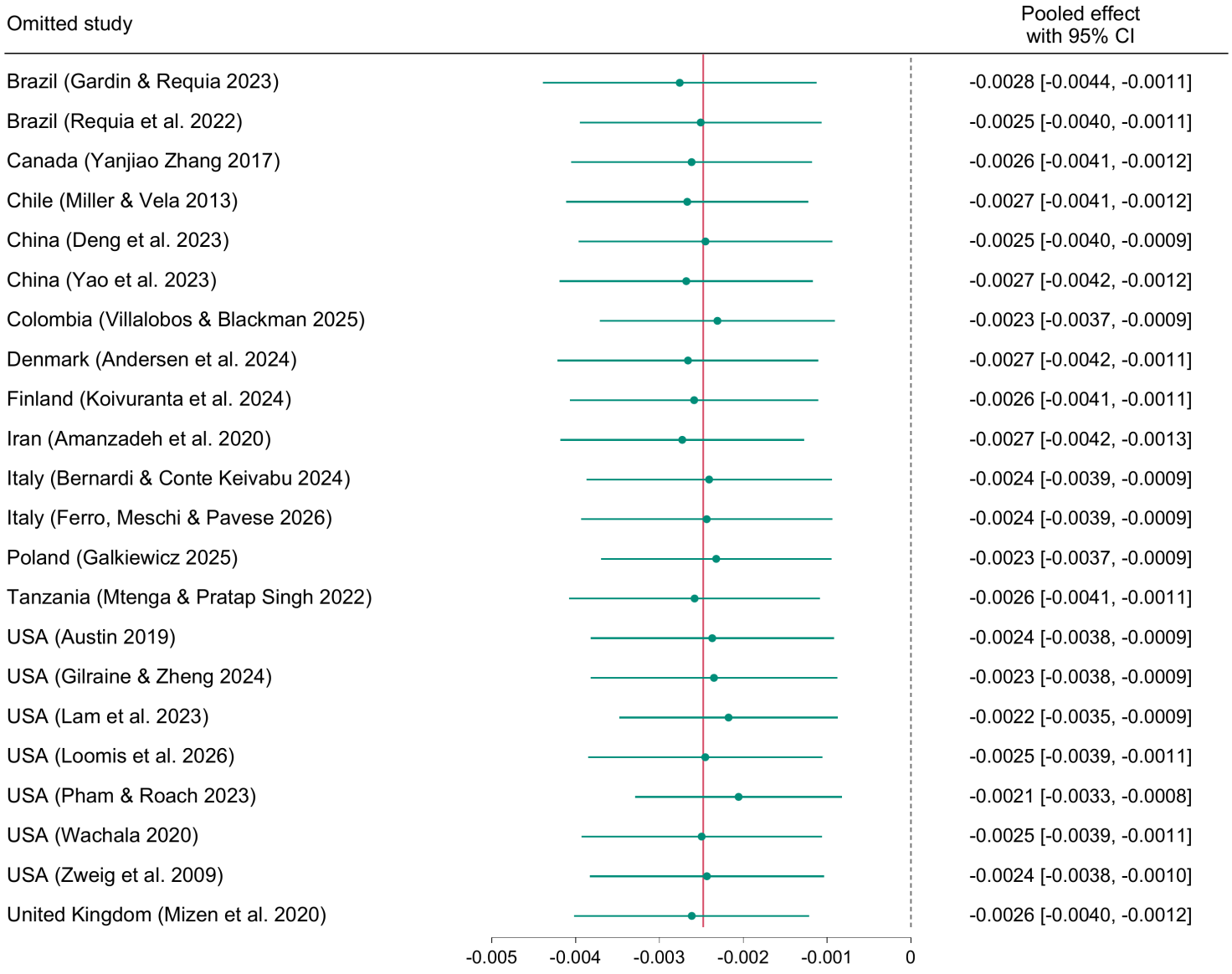
Which conclusions depend on empirical choices?

Restriction	$PM_{2.5}$	Heat
All eligible	-0.0025	-0.0089
95% CI	[-0.0039, -0.0011]	[-0.0160, -0.0019]
Individual FE	-0.0019	-0.0098
95% CI	[-0.0035, -0.0004]	[-0.0212, 0.0015]
Exclude group SDs	-0.0025	-0.0089
95% CI	[-0.0040, -0.0010]	[-0.0160, -0.0019]
Also exclude unresolved	-0.0023	-0.0063
95% CI	[-0.0037, -0.0008]	[-0.0120, -0.0006]

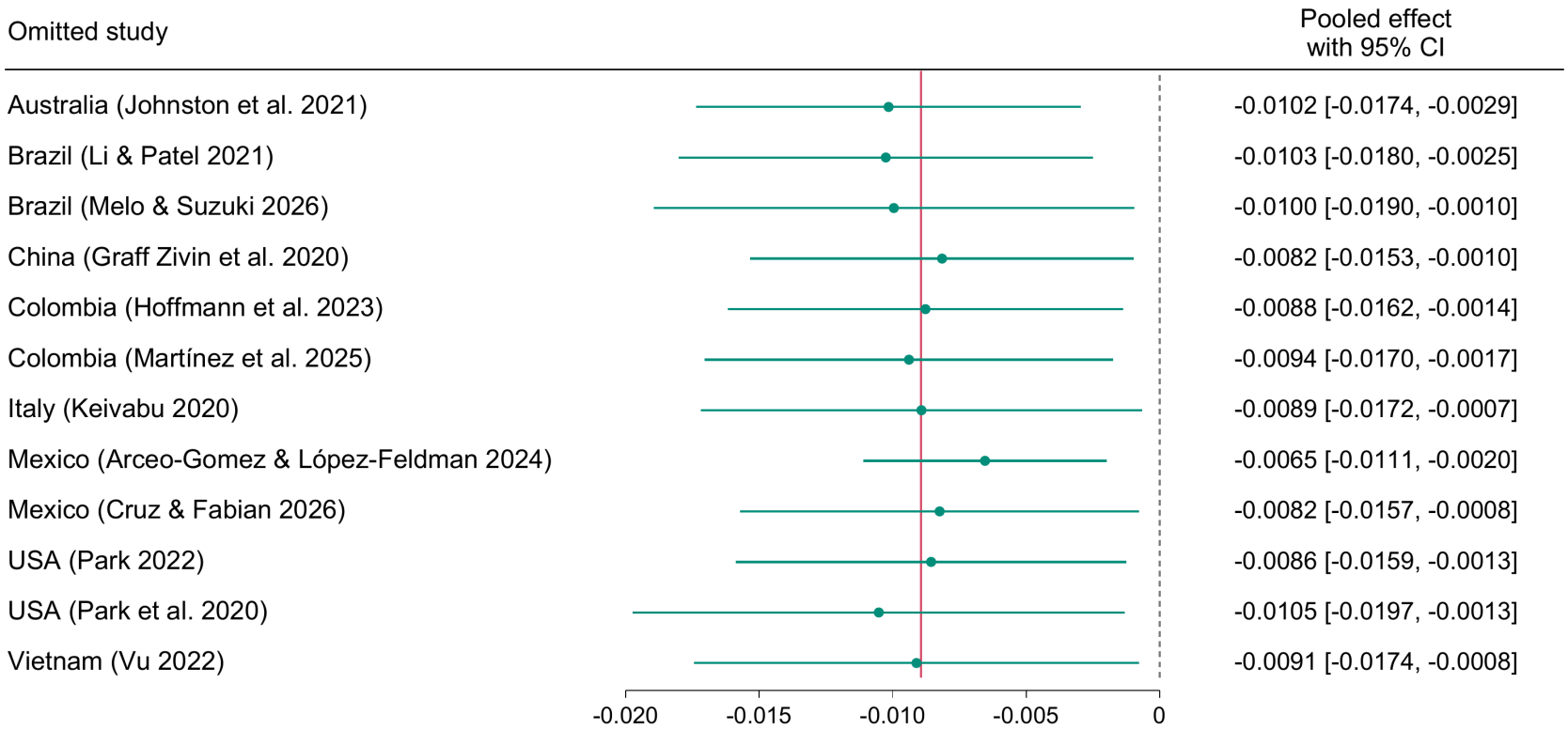
Tables A6–A7: multiple-estimate per-unit effects. Group SDs refer to group-average scores. The last row additionally excludes unresolved normalization cases.

$PM_{2.5}$ remains negative across these checks; the individual-FE heat interval includes zero. Restrictions change sample composition as well as precision.

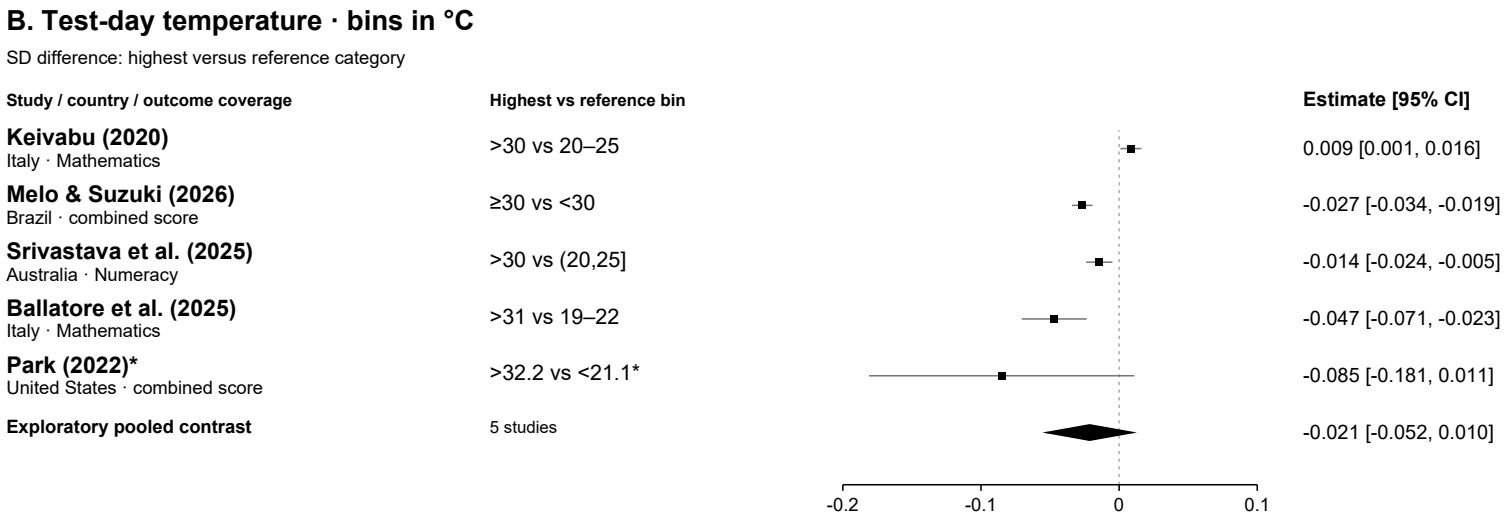
Leaving out one $PM_{2.5}$ study



Leaving out one heat study



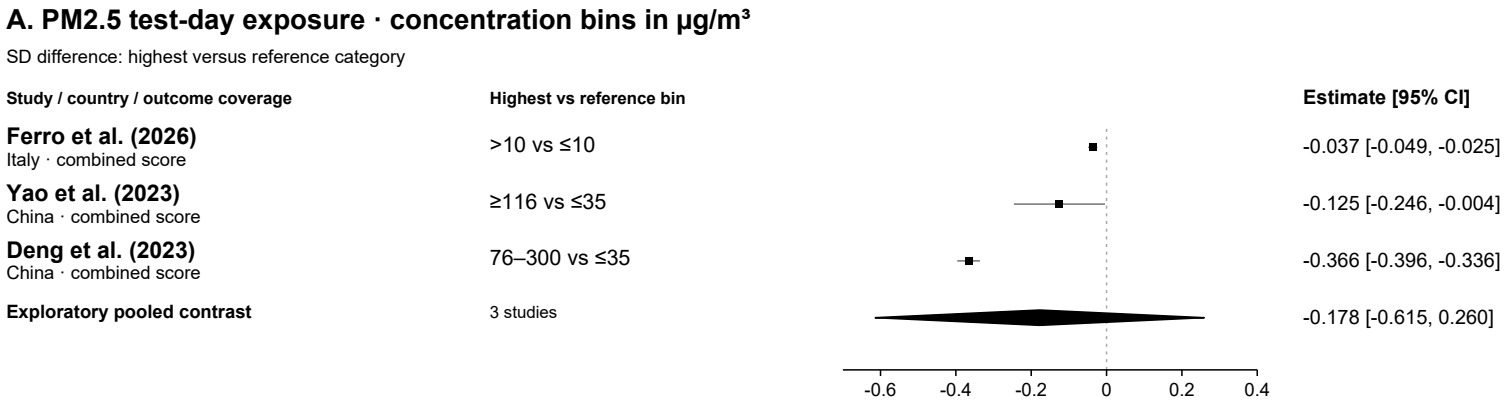
Nonlinear effects: heat on test day



Notes: Effects in test-score SD; 95% CIs. Park's reference bin is assumed because the source gives conflicting definitions.

- **Interpretation:** Taking the test in the hottest bin rather than the study's reference temperature bin.
- **Finding:** The pooled estimate is negative, but its confidence interval includes zero; temperature thresholds differ across studies.

Nonlinear effects: pollution on test day



Notes: Effects in test-score SD; 95% CIs. $PM_{2.5}$ thresholds, reference categories, and study designs differ. Exposure windows range from examination hours to the preceding 24 hours.

- **Interpretation:** Compare scores under high pollution with scores in the study's reference exposure category.
- **Finding:** All three estimates are negative, but the pooled confidence interval includes zero.

Pollution exposure profiles

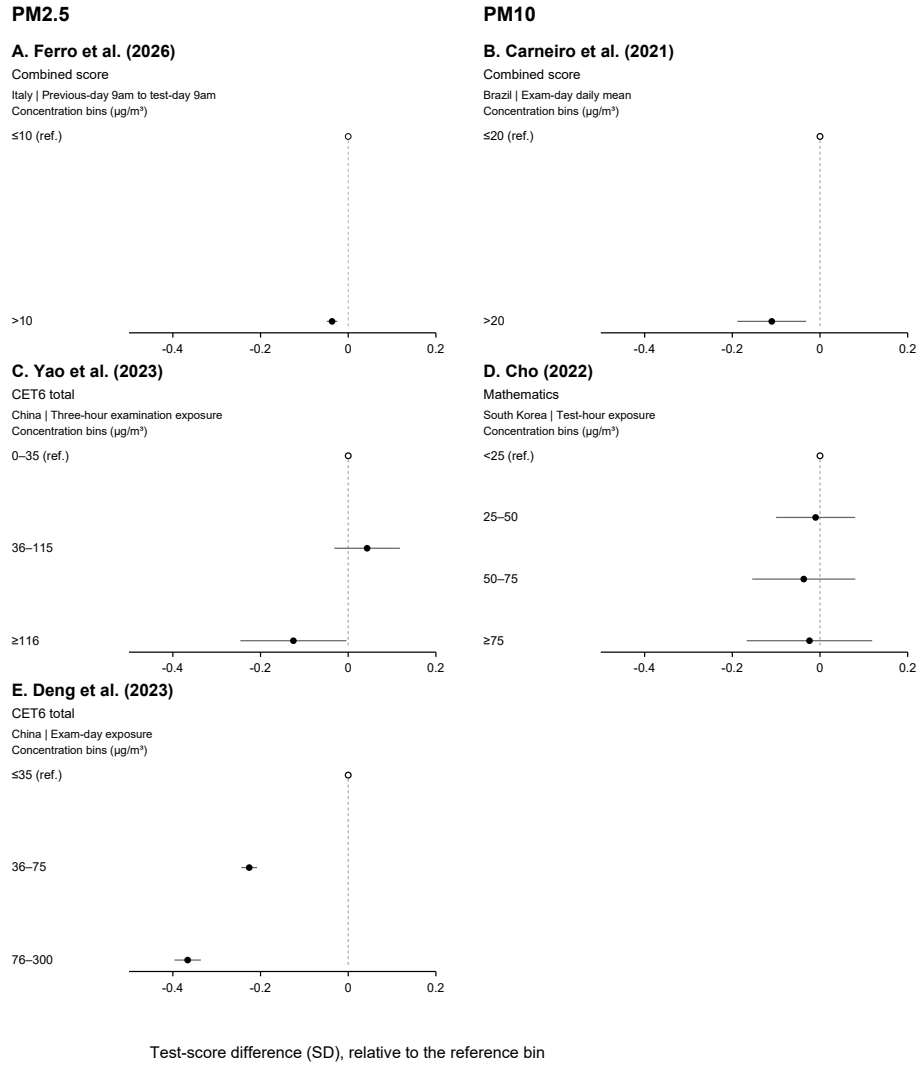


Figure A10. Study-specific categories; spacing is categorical, not exposure distance. Bars are pointwise 95% CIs.